STANPUMP CET Simulation Report

Propofol TCI — Effect-Site Targeting across 7 Patients and 4 PK/PD Models

1. Background: Propofol Pharmacokinetics and Pharmacodynamics

Propofol is the primary intravenous anaesthetic used for total intravenous anaesthesia (TIVA). Its effect — loss of consciousness and anaesthetic depth — is mediated at the brain, not in the blood. Because propofol distributes from blood into the brain (the effect site) with a finite rate constant (ke0), plasma concentration (Cp) and effect-site concentration (Ce) are not equal during dynamic dosing. Targeting Cp directly overshoots during induction and undershoots during wake-up. Effect-site targeting (CET) accounts for this lag.

Three-compartment model — propofol pharmacokinetics are described by:

Compartment	Physiology	Linked by
1 (central)	Blood / highly perfused organs	k12, k21, k13, k31
2 (fast peripheral)	Muscle	k12, k21
3 (slow peripheral)	Fat	k13, k31
Effect site	**Brain**	**ke0 (= k41)**

Drug is eliminated from compartment 1 at rate k10. Amounts in each compartment follow a system of linear ODEs whose solution is a sum of exponentials — the mathematical structure STANPUMP exploits.

2. The STANPUMP Algorithm

STANPUMP (Shafer & Gregg 1992) is the reference implementation for computer-controlled infusion in anaesthesia. It operates in **discrete 1-second steps** — confirmed from SimTIVA source (`refresh_interval = 1000 ms`, `delta_seconds = 1`) — and uses closed-form exponential recursion rather than numerical ODE integration, giving exact results at machine speed.

2.1 Characteristic Roots — `cube()`

For a 3-compartment model the characteristic polynomial is:

```
X³ + a₂X² + a₁X + a₀ = 0

  a₀ = k10·k21·k31
  a₁ = k10·k31 + k21·k31 + k21·k13 + k10·k21 + k31·k12
  a₂ = k10 + k12 + k13 + k21 + k31
```

The polynomial is transformed to depressed form `y³ + p·y + q = 0` and solved analytically via the trigonometric (Cardano) method. All three roots are real and negative for a physically valid model. The roots $\lambda_1 \leq \lambda_2 \leq \lambda_3$ are the exponential decay rates. A fourth root λ_4 = ke0 describes effect-site equilibration.

2.2 Macro-Constants — `calculate_udfs()`

Plasma coefficients (p_coef) link infusion rate to Cp:

```
p_coef[i] = (k21 − λᵢ)(k31 − λᵢ) / [(λᵢ − λⱼ)(λᵢ − λₖ) · Vc · λᵢ]
```

Effect-site coefficients (e_coef):

```
e_coef[i] = p_coef[i] · ke0 / (ke0 − λᵢ)          i = 1,2,3
e_coef[4] = (ke0 − k21)(ke0 − k31) / [(λ₁ − ke0)(λ₂ − ke0)(λ₃ − ke0) · Vc]
```

Unit Disposition Functions (UDF) capture the response to a unit infusion (1 mg/s):

```
p_udf[t] = Cp from 1 mg/s continuous infusion running for t seconds  (tabulated 0..300 s)
e_udf[t] = Ce at time t following 1 mg/s infusion for 1 second then off
```

UDFs are computed by the exact 1-second recursion:

```
state[i](t+1) = state[i](t) · exp(−λᵢ) + coef[i] · rate · (1 − exp(−λᵢ))
```

`peak_time` is the index at which e_udf reaches its maximum — the moment Ce peaks after a unit bolus.

2.3 Effect-Site Targeting (CET) — Induction

Starting from zero drug in the system, the initial bolus is determined by:

1. **First estimate:** `trial_rate = Ce_target / e_udf[peak_time]`
2. **Iterative refinement** via `find_peak()`: hill-climb to find the time that maximises `Ce_from_existing_state + e_udf[t] · trial_rate`, then recompute `trial_rate`. Repeat until convergence (|achieved − target| < 0.01 %).
3. **Round up:** `bolus_mg = ceil(trial_rate)` — always overshoot slightly at peak to avoid under-dosing.
4. **Administer bolus** in 1 second; hold pump off until `peak_time` seconds have elapsed.

2.4 Plasma Targeting (CPT) — Maintenance

After the induction peak, STANPUMP switches to plasma-concentration targeting (CPT). Because at steady state Cp ≈ Ce, maintaining Cp at the target keeps Ce there too.

Each second (δ = 1 s):

```
predicted_Cp = Σᵢ ps[i] · exp(−λᵢ · δ)     (Cp in 1 s if pump is off)
rate = max(0,  (Ce_target − predicted_Cp) / p_udf[δ])
```

2.5 Per-Second State Update

All compartmental dynamics reduce to seven scalar recursions:

```
# Plasma  (l[i] = exp(−λᵢ · 1s))
ps[i] = ps[i] * l[i] + p_coef[i] * rate * (1 − l[i])     i = 0,1,2

# Effect site  (l[3] = exp(−ke0 · 1s))
es[j] = es[j] * l[j] + e_coef[j] * rate * (1 − l[j])     j = 0,1,2,3
```

`Cp = Σ ps[i]`, `Ce = Σ es[j]`.

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3. PK/PD Models Used

Model	Reference	Validated age range	Key characteristic
Marsh	BJA 1991;67:41–8	Adults	Weight-only scaling; Vc = 0.228·BW; all k fixed
Schnider	Anesthesiology 1998;88:1170–82	Adults	Fixed Vc = 4.27 L; CL1 adjusted for LBM, age, height
Paedfusor	BJA 2003;91:507–13	1–16 yr → Adults	Age-banded k10 and Vc; ke0 from Tpeak formula (BJA 2008)
Eleveld	**BJA 2018;120:942–59**	**Neonates → Elderly**	**Allometric + maturation functions; universal model**

All models return rate constants in min^{-1} and volumes in litres; STANPUMP converts to per-second internally.

3.1 Applicability to Paediatric Patients

- **Marsh:** designed for adults; k-rates are weight-independent — underestimates requirements for young children and does not adapt ke0 to age.
- **Schnider:** fixed Vc (4.27 L) is unrealistic for small children; ke0 (0.456 min^{-1}) not age-appropriate.

- **Paedfusor:** purpose-built for children; switches to adult (Marsh-equivalent) parameters at age ≥ 16 yr.
- **Eleveld:** fully continuous across ages via maturation functions; the only model simultaneously valid for all seven patients.

4. Simulation Protocol

Parameter	Value
Ce target	4 µg/mL
Procedure duration	90 min (5 400 s)
Syringe content	50 mL × 10 mg/mL (1% propofol) = **500 mg**
Syringe change — 1st	20 s
Syringe change — 2nd	**180 s** (prolonged — stress-test scenario)
Syringe change — 3rd+	20 s
STANPUMP time step	1 s (confirmed from SimTIVA source)
CPT rate update	**every 1 s**

The 180-second second change is the scenario's stress test: a 3-minute interruption mid-procedure forces Ce to drift below target before the pump recovers.

5. Patients

#	Sex	Age	Weight	Height	BMI
1	Male	5 yr	18 kg	110 cm	14.9
2	Female	10 yr	30 kg	140 cm	15.3
3	Male	20 yr	80 kg	180 cm	24.7
4	Female	20 yr	50 kg	150 cm	22.2
5	Male	40 yr	80 kg	180 cm	24.7
6	Female	40 yr	50 kg	150 cm	22.2
7	**Male**	**40 yr**	**120 kg**	**180 cm**	**37.0**

Patients 1–2 are paediatric; patients 3–7 are adults spanning sex, BMI, and the 40-year working age group. Patient 7 is obese (BMI 37 kg/m²).

6. Results

6.1 Summary Figure

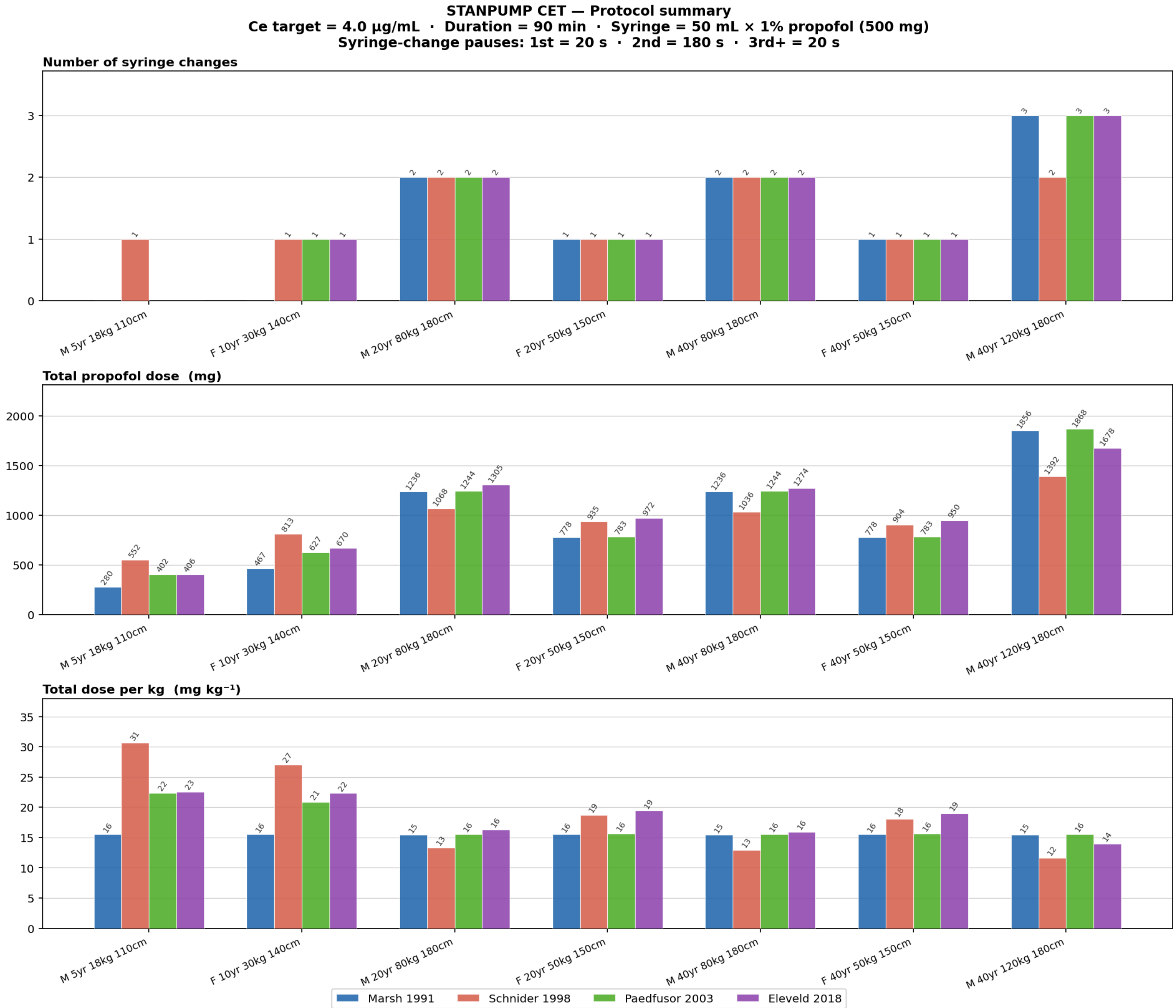

Figure 1. Summary across all 7 patients and 4 models. Top: number of syringe changes. Middle: total propofol dose (mg). Bottom: dose per kg body weight (mg/kg).

Three stacked panels for all 7 patients × 4 models.

Panel 1 — Syringe changes. Small children (patients 1–2) typically complete the procedure on one syringe. Large adults (80–120 kg) require 2–3 changes. Schnider consistently requires one fewer change for large patients because its fixed small Vc leads to lower maintenance rates once equilibrium is reached.

Panel 2 — Total dose (mg). Absolute dose scales with body weight. Marsh and Paedfusor (for adults) are nearly identical and purely weight-proportional. Schnider shows less weight-proportionality (fixed Vc, LBM-adjusted clearance). Eleveld gives the largest boluses (longest peak_time) but comparable total dose for most adults.

Panel 3 — Dose per kg (mg/kg). Normalised dose reveals model philosophy most clearly: Marsh is flat ~15.5 mg/kg for all adults. Schnider drops sharply for the obese patient 7 (11.6 mg/kg) — effectively dose-capping in obesity. Paedfusor gives children more per kg (15.5–22 mg/kg), which is physiologically correct. Eleveld (15–22 mg/kg) applies sex and age corrections; females receive slightly more per kg due to larger CL1.

6.2 Patient Panels

Each figure shows a 2×2 grid (one panel per model). Within each panel:

Element	Description
Blue solid line	Effect-site concentration Ce (µg/mL), left axis
Orange dashed line	Plasma concentration Cp (µg/mL), left axis
Black dashed horizontal	Target Ce = 4 µg/mL
Gray filled area	Infusion rate (ml/h), right axis — scaled to maintenance rates; bolus spike may clip
Orange bands	Syringe-change pauses, labelled with duration in seconds
Annotation box	**Initial bolus (mg), peak time (s), total dose, number of changes**

All four panels share the same concentration y-axis, scaled to 120% of the 99.5th-percentile Cp across all models — zooming in on maintenance while avoiding compression by the brief induction Cp spike.

Patient 1 — Male, 5 yr, 18 kg, 110 cm

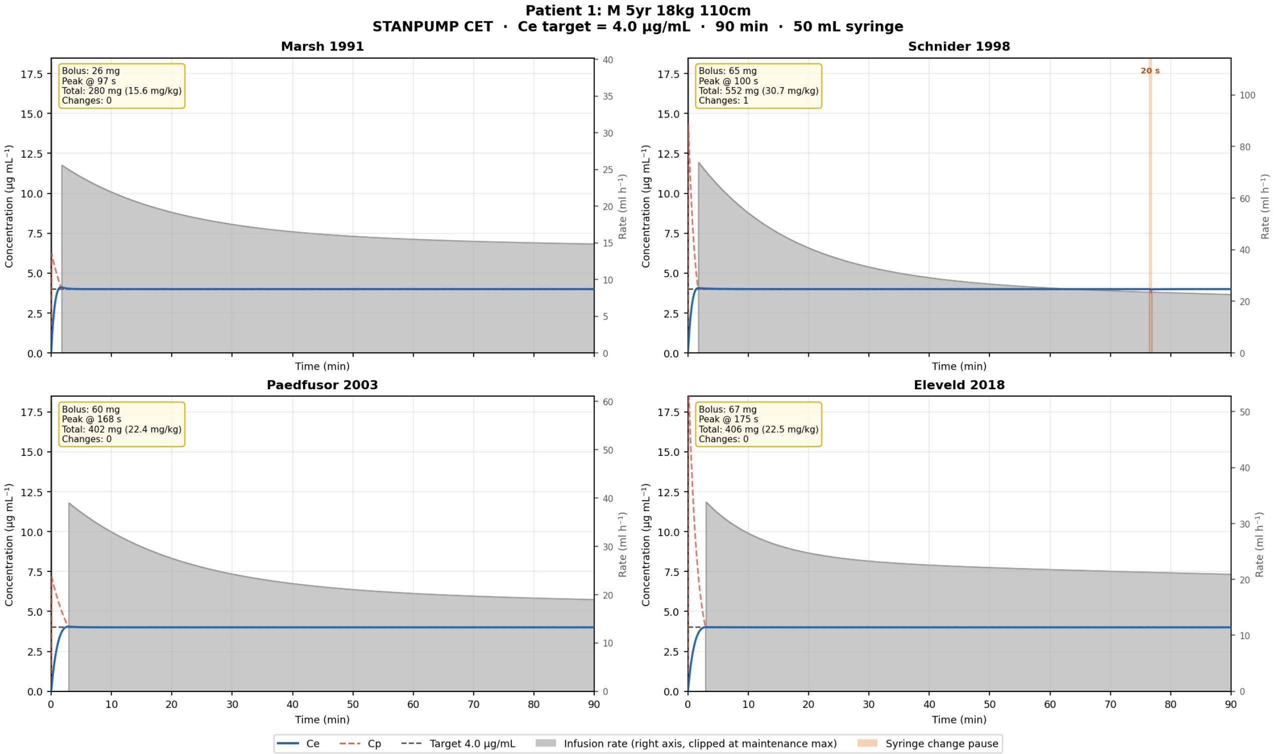

Figure 2. Patient 1 (M 5yr 18kg 110cm) — all four models.

Clearest model divergence in the dataset. **Marsh** delivers only a 26-mg bolus (flat weight scaling) with adult rate constants — Ce reaches target, but the underlying model is biologically inappropriate for a 5-year-old. **Paedfusor** and **Eleveld** use a longer peak_time (~168–175 s) and larger bolus (60–67 mg), correctly reflecting the child's slower ke0 and higher volume of distribution per kilogram. **Schnider** produces the highest mg/kg dose (30.7 mg/kg) because its adult-fixed Vc of 4.27 L is disproportionately small for this child.

Patient 2 — Female, 10 yr, 30 kg, 140 cm

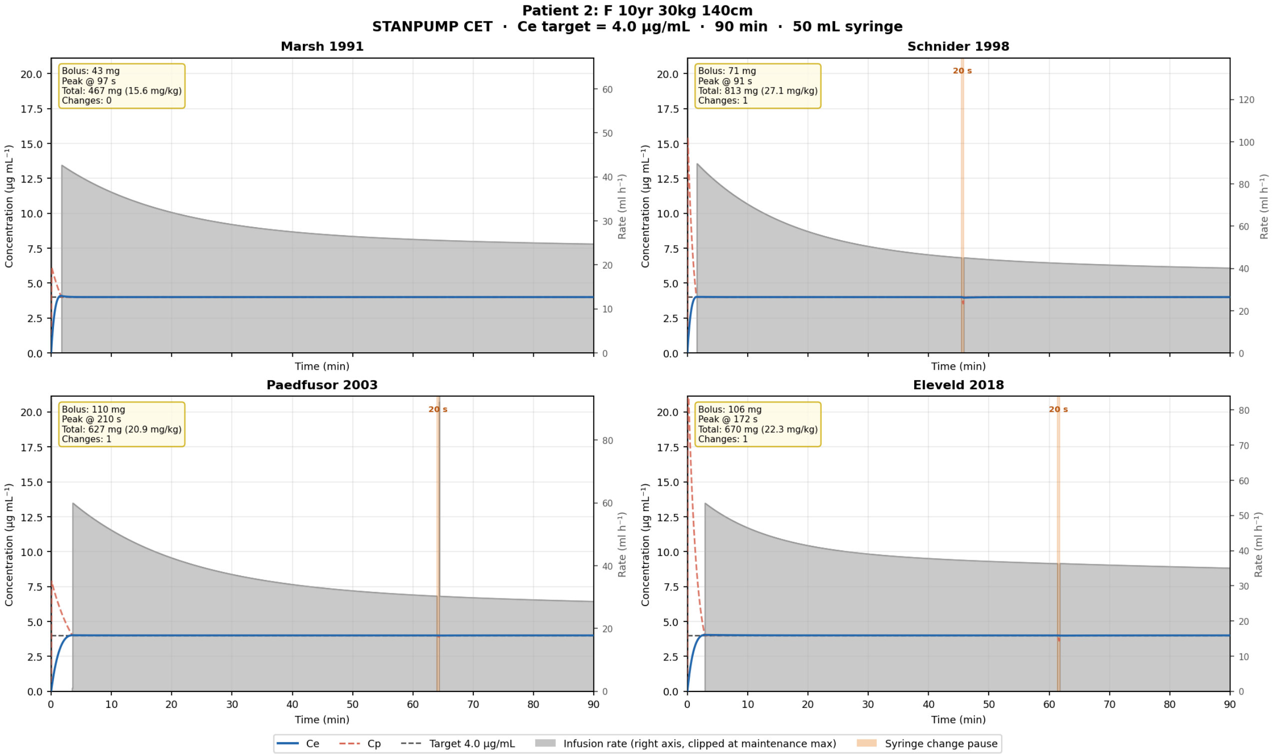

Figure 3. Patient 2 (F 10yr 30kg 140cm) — all four models.

Paedfusor produces the longest induction pause (210 s) — its age-dependent ke0 formula gives the slowest equilibration at 10 yr, consistent with published Tpeak data. **Eleveld** peak_time (172 s) is shorter but still clearly paediatric. **Marsh** significantly under-doses per kg (15.6 mg/kg vs expected ~20 mg/kg for this age).

Patient 3 — Male, 20 yr, 80 kg, 180 cm

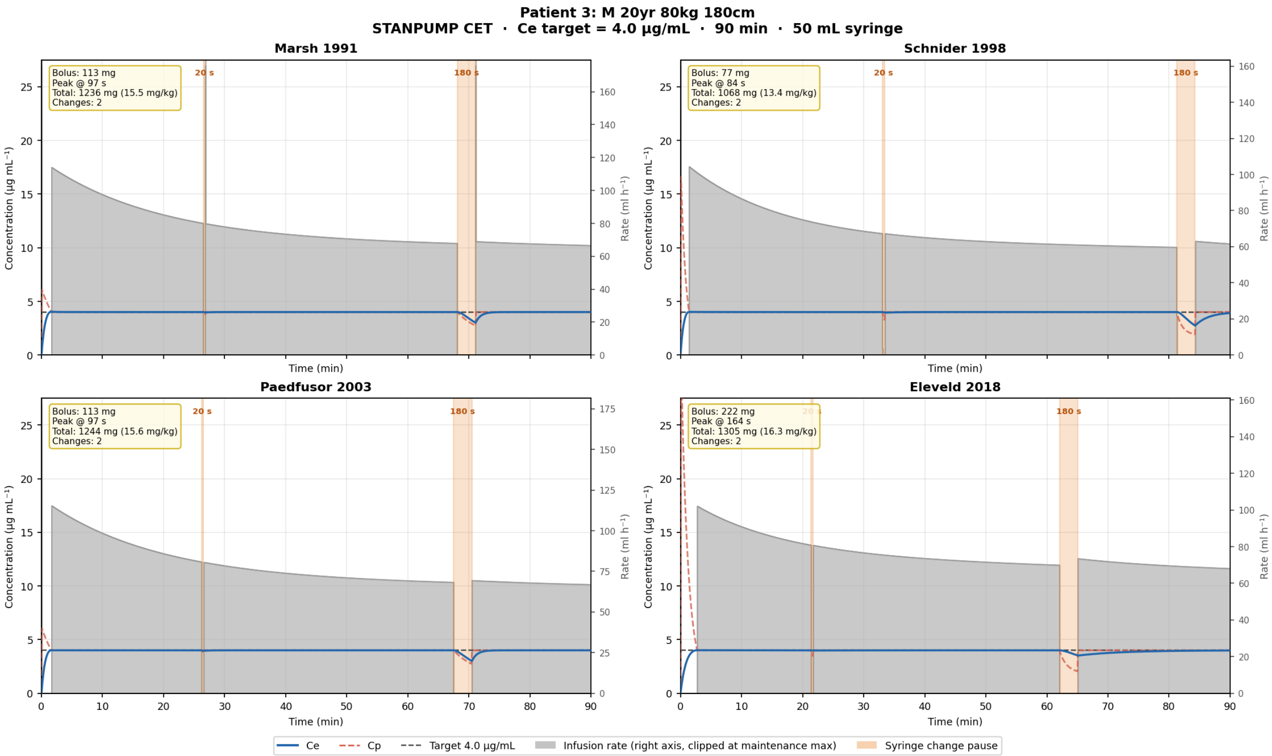

Figure 4. Patient 3 (M 20yr 80kg 180cm) — all four models.

Adult models begin to agree more closely. **Schnider**'s fixed Vc creates a higher initial Cp spike that decays quickly. **Eleveld** requires the largest bolus (222 mg) due to its longer peak_time (164 s). Two syringe changes required by all models; the 180-second second change produces a visible Ce dip below target.

Patient 4 — Female, 20 yr, 50 kg, 150 cm

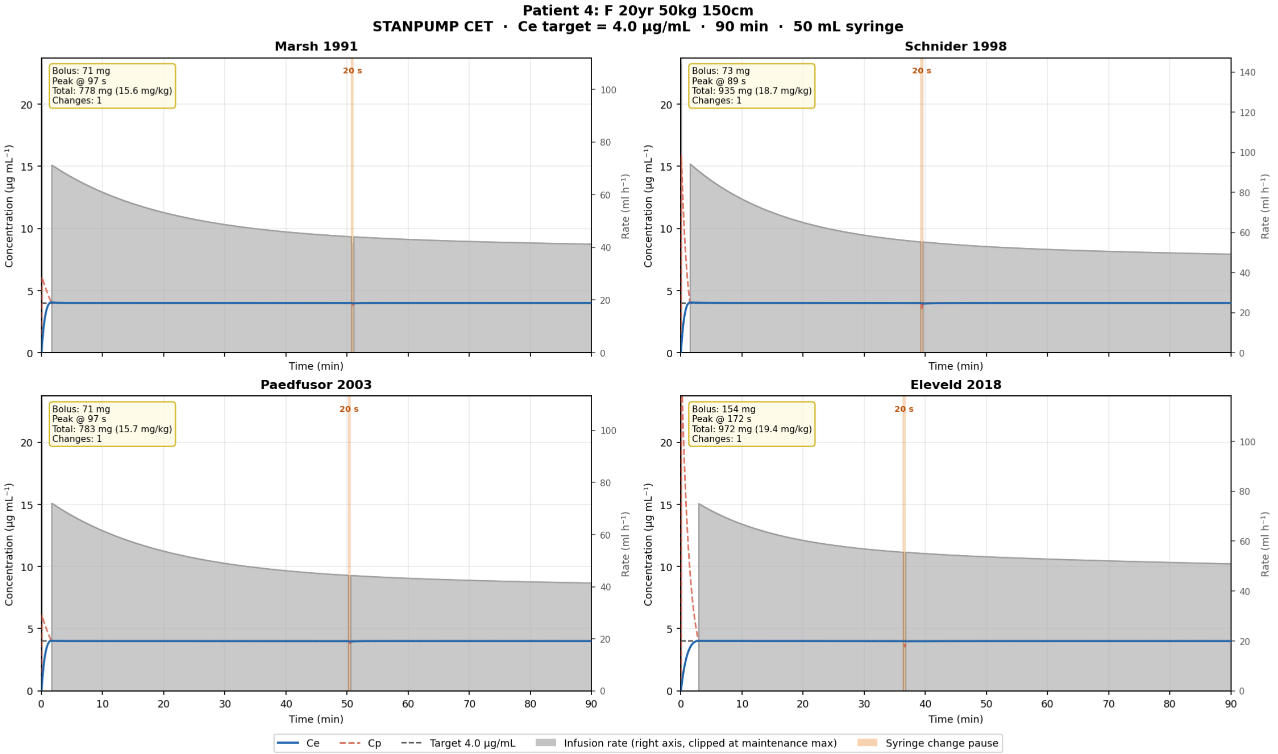

Figure 5. Patient 4 (F 20yr 50kg 150cm) — all four models.

Sex effect visible in **Eleveld**: female CL1 is 2.1 L/min vs 1.79 L/min for males, giving a larger bolus (154 mg) and higher total dose (19.4 mg/kg) than the equivalent male. **Schnider** also gives higher doses for females through LBM correction. One syringe change for all models.

Patient 5 — Male, 40 yr, 80 kg, 180 cm

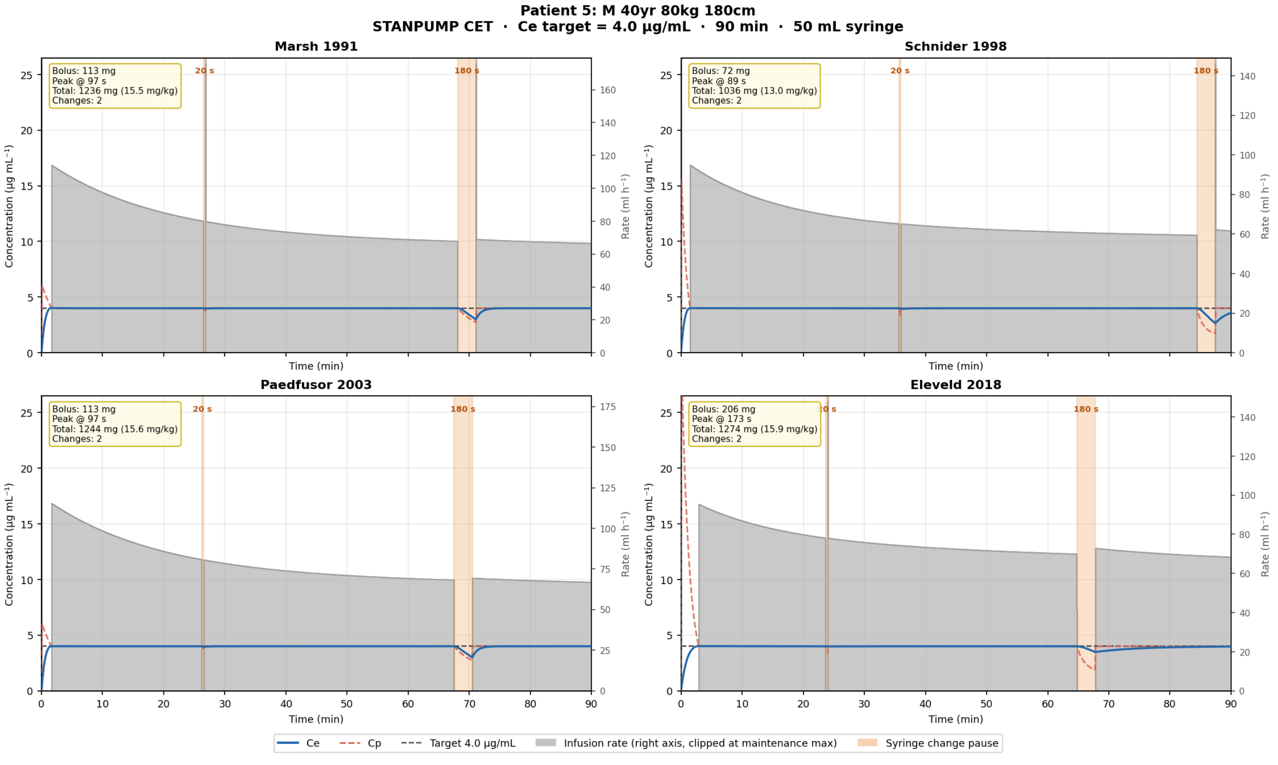

Figure 6. Patient 5 (M 40yr 80kg 180cm) — all four models.

Closely matches patient 3 (same weight/height, 20 years older). The 180-second change is the second change here, producing the deepest Ce dip across all models. Slight age-dependent clearance decline visible in **Eleveld**.

Patient 6 — Female, 40 yr, 50 kg, 150 cm

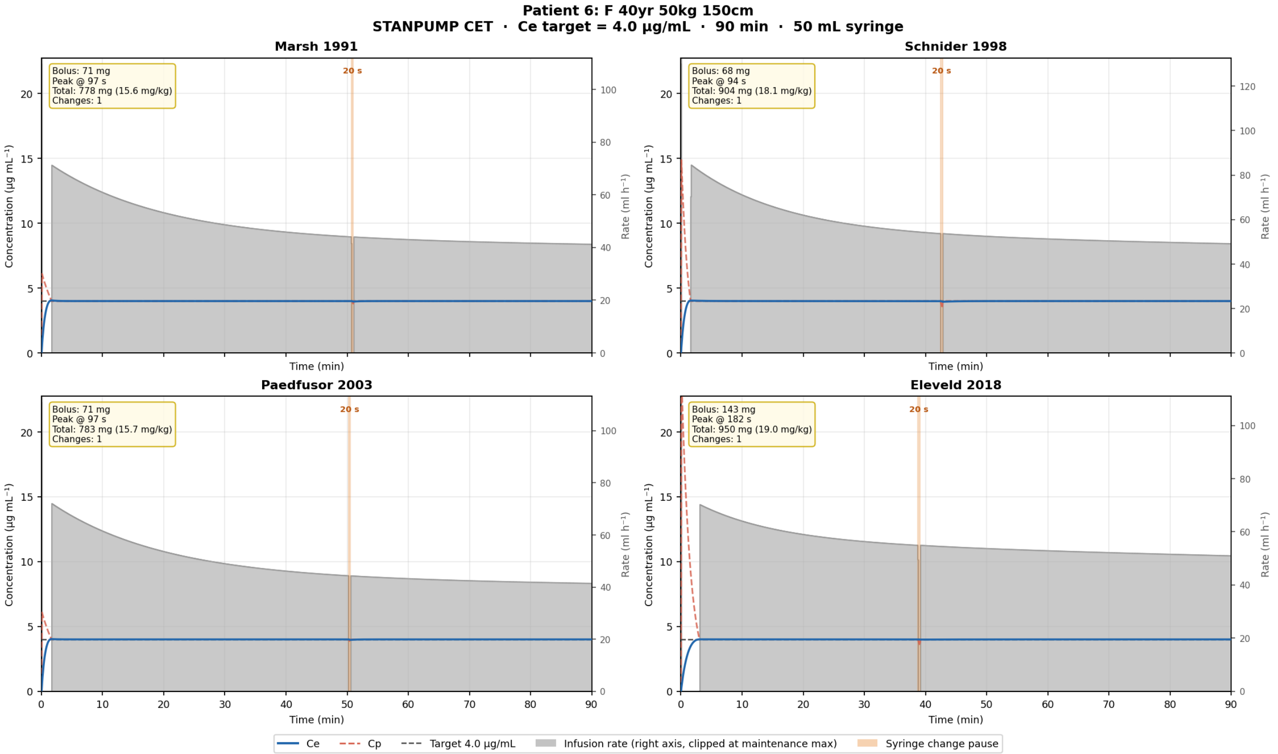

Figure 7. Patient 6 (F 40yr 50kg 150cm) — all four models.

Female patients consistently require higher mg/kg doses under **Eleveld** and **Schnider** compared to weight-matched males. **Marsh** and **Paedfusor** (adult parameters) are sex-blind and give 15.7 mg/kg regardless. The prolonged 180-second change produces a clear Ce drop and recovery ramp visible in the blue Ce line.

Patient 7 — Male, 40 yr, 120 kg, 180 cm (obese, BMI 37)

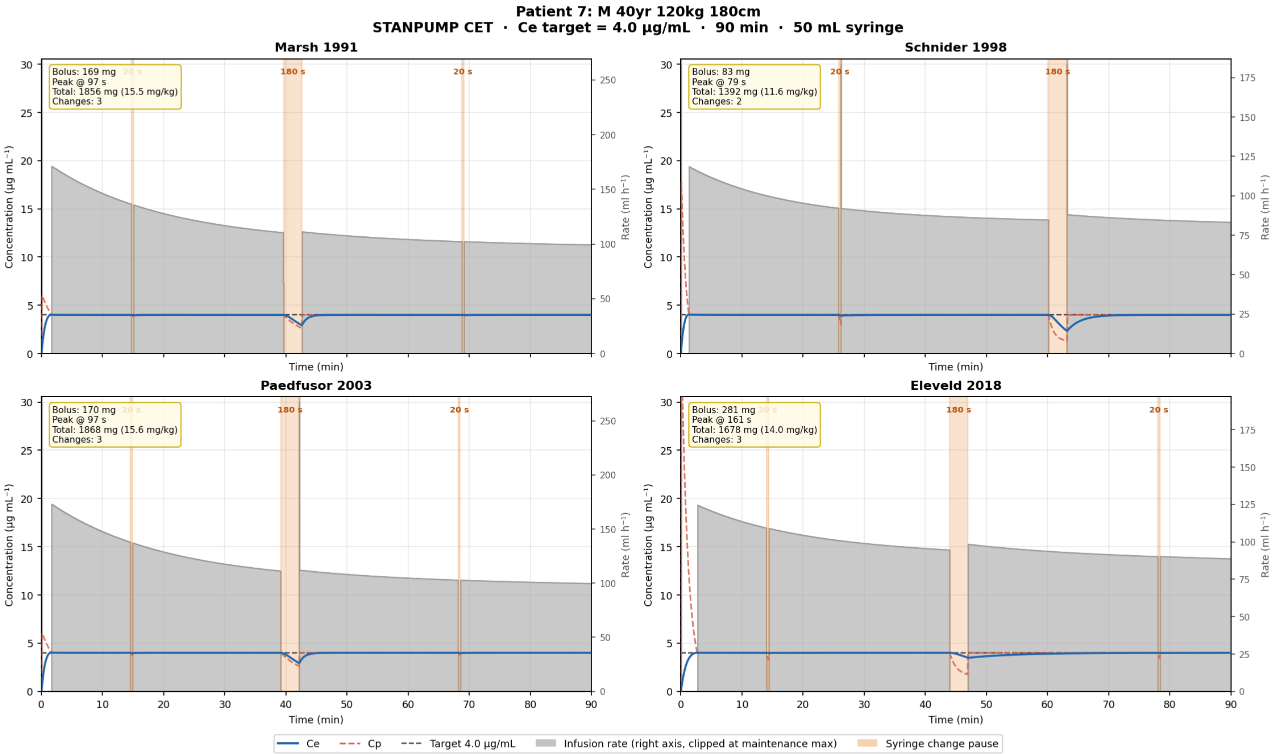

Figure 8. Patient 7 (M 40yr 120kg 180cm) — all four models.

Greatest divergence among adult patients.

- **Marsh / Paedfusor:** 169–170 mg bolus; three syringe changes; flat 15.5 mg/kg — obesity ignored completely.
- **Schnider:** 83-mg bolus (smallest) and lowest total dose (11.6 mg/kg) — LBM-based clearance with fixed Vc corrects aggressively downward for obesity. Requires only 2 changes.
- **Eleveld:** 281-mg bolus (largest); allometric weight scaling increases both Vc and CL1, giving a moderate obesity correction (14.0 mg/kg) that is physiologically more grounded than Schnider. Still 3 changes.

7. Key Observations

1. **STANPUMP runs at 1-second resolution** — confirmed from SimTIVA source (`refresh_interval = 1000 ms`, `delta_seconds = 1`). Both the UDF tables and the CPT rate update operate at this cadence.
2. **Model choice matters most for children** — Marsh and Schnider give biologically implausible dosing for patients under 13 yr. Only Paedfusor and Eleveld are appropriate for paediatric use.
3. **Schnider most aggressively corrects for obesity** — fixed Vc with LBM-based clearance gives the lowest mg/kg dose in patient 7. Eleveld's allometric approach is more physiologically grounded.
4. **The 180-second change causes clinically significant Ce drop** — visible as an orange band followed by a Ce dip in every affected panel. Recovery time depends on model k-rates and residual drug burden in peripheral compartments.
5. **Eleveld uses larger initial boluses** — because ke0 is smaller (longer time to peak), more drug must be front-loaded. Total 90-minute dose remains comparable because maintenance rate is then lower.
6. **Syringe change count is a clean proxy for total drug load** — each additional change represents approximately 500 mg more propofol.

8. Implementation Notes

File	Purpose
`stanpump_tci.py`	Core STANPUMP engine: `cube()`, `calculate_udfs()`, `virtual_model_cp/ce()`, `find_peak_time()`, `simulate_stanpump_cet()`
`stanpump_scenario.py`	Protocol constants, model instantiation, `run_all()`
`stanpump_plots.py`	All figures: summary (3-panel bar) + per-patient (2×2 grid)
`models/marsh.py`	Marsh model (BJA 1991)
`models/schnider.py`	Schnider model (Anesthesiology 1998)
`models/paedfusor.py`	Paedfusor model (BJA 2003)
`models/eleveld.py`	**Eleveld model (BJA 2018)**

Reproduce all results:

```
cd PKPD_Reimplementation
python3 stanpump_plots.py     # simulates all patients × models and saves all figures
```

9. References

1. Shafer SL, Gregg KM. Algorithms to rapidly achieve and maintain stable drug concentrations at the site of drug effect with a computer-controlled infusion pump. J Pharmacokinet Biopharm 1992;**20**:147–69.
2. Marsh B, et al. Pharmacokinetic model driven infusion of propofol in children. Br J Anaesth 1991; **67**:41–8.

3. Schnider TW, et al. The influence of method of administration and covariates on the pharmacokinetics of propofol in adult volunteers. Anesthesiology 1998;**88**:1170–82.
4. Absalom A, et al. Pharmacokinetic model for propofol for use with a target-controlled infusion system in paediatric patients (Paedfusor). Br J Anaesth 2003;**91**:507–13.
5. Eleveld DJ, et al. Pharmacokinetic–pharmacodynamic model for propofol for broad application in anaesthesia and sedation. Br J Anaesth 2018;**120**:942–59.
6. Jeleazcov C, et al. Pharmacodynamic modelling of the bispectral index response to propofol-based anaesthesia in children. Br J Anaesth 2008;**100**:509–16.
7. SimTIVA open-source TCI simulator — pharmacology.js (source of the STANPUMP JavaScript implementation translated into Python here).